

1. Abstract

This project uses a smaller version of Zillow's housing dataset to predict each property's total assessed value ('taxvaluedollarcnt', land plus structure, in U.S. dollars). Our objective for this project was to build an accurate model that generalizes well with low prediction error based on the dataset. Our approach began with performing exploratory data analysis (EDA) to assess the quality of our data, understand the feature distributions, identify missing data, and determine variable types. We found that two features were right skewed, and so we applied log transformations to stabilize their distributions and reduce outlier influence; we also created a few simple ratio features. The dataset was split into training and test sets, and model selection was done with cross-validation configured to minimize mean absolute error. We trained and compared three models: GradientBoostingRegressor (GBR), Random Forest, and Decision Tree, with modest grid searches for key hyperparameters. On the held-out test set, GradientBoostingRegressor achieved the lowest MAE and produced the most stable errors across value ranges. Tree-based feature importance and wrapper methods consistently retained the engineered features, indicating they added useful signal. The final deliverable is a trained gradient boosting model and reproducible code covering preprocessing, feature engineering, tuning, and evaluation.

2. Introduction

In the real estate market, accurately predicting the total valuation of a property is vital for providing reasonable selling prices, fair property tax assessments, and better informed investment decisions. Using a subset of the Zillow housing data, our project aims to predict total property assessed values by developing a machine learning regression model. The problem we are specifically addressing is to build a predictive model that will accurately forecast the

dataset's target, 'taxvaluedollarcnt', the total assessed value of a property in U.S. dollars that combines land and structure. Valuation is challenging because public records can be incomplete or inconsistent, predictors mix numeric and categorical fields, and relationships with value are often nonlinear and location-dependent, which can produce biased or unstable estimates. Thus, this task also involves handling data that consists of missing values, redundant features, skewed feature distributions, and outliers, which, if left unresolved, can affect the predictability of the model. Our objectives include performing a preliminary exploratory data analysis to better understand the features, cleaning and preprocessing the data to address any flaws, feature engineering and transformation to uncover underlying patterns, training and evaluating three machine learning regression models, and optimizing these models through feature selection and hyperparameter tuning. Ultimately, with this, our goal is to provide the model that performs the best in terms of predictive accuracy and reliability, ensuring consistent results and meaningful insights into property valuation.

3. Data Description

The dataset used in this project is from the Zillow Million Dollar Prize hosted on Kaggle, provided by Zillow. We worked with a smaller subset that contains 77,414 rows and 55 columns. The dataset includes a variety of housing property descriptors such as property attributes, location details, amenities, and assessment information. The target variable is 'taxvaluedollarcnt', the total assessed value of a property in U.S. dollars that combines land and structure. Key predictors include 'bathroomcnt', 'calculatedfinishedsquarefeet', 'roomcnt', and 'yearbuilt'. Most features are numeric floats, with a few categorical fields stored as objects. Except for the 'parcelid', which doesn't have missing values, all the other features contain missing values. The highest missing percent is 99.98% and the lowest is 0.04%. Some features in

the dataset are redundant or contain many missing values. Also, samples from the target variable, `taxvaluedollarcnt`, include many nulls or outliers, which were addressed in the data preprocessing steps. This dataset is complex and offers enough variability for training and evaluating regression models for predicting the target.

4. Methodology

The overall analytical framework involves understanding the dataset through exploratory data analysis, uncovering patterns, and gathering meaningful insights for modeling. This involves systematic steps in data cleaning and preprocessing, feature engineering and selection, model selection and evaluation, and hyperparameter tuning, which all help to develop a robust and reliable predictive regression model. We applied the `GridSearchCV` to search for the optimal hyperparameters instead of using the validation curves, as it can be done in a single step, despite it taking longer than the validation curves. We selected Mean Absolute Error (MAE) as the primary evaluation metric because it is not overly affected by extreme outliers and it uses the same unit as the target variable, which is in dollars, making it more readable and less sensitive compared to MSE.

During the data cleaning and preprocessing steps, we referred to Appendix 1 in our project to understand the features we were working with to get an overall better grasp of the dataset. After a preliminary EDA, which involved using methods like `.info()`, `.describe()`, `.nunique()`, and plotting histograms for visualizations, we decided to drop some features that were not suitable for our regression task. We removed features that were redundant, had similar definitions to other features, and contained many missing values, such as 'parcelid', 'taxdelinquencyyear', 'finishedsquarefeet13', and sixteen other features. Also, samples that contained many nulls or outliers were dropped from the target variable,

‘taxvaluedollarcnt’, using the IQR method. To avoid a significant loss of data, some features with fewer missing values were handled using imputation methods, such as median, mode, and replacing and filling in with 0. Encoding was performed on both object and category features to better prepare the data. Additionally, part of our EDA involved investigating feature relationships to get a thorough understanding of our dataset. This involved computing the correlation matrix, plotting a correlation heatmap, computing the F-statistic, and exploring forward and backward feature selection. This helped us to identify which features strongly correlated with the target variable and which were the most relevant for our regression model. From this, what worked was cleaning the data without losing significant data. We were able to encode categorical features and analyze feature relationships to determine the most valuable features. What didn’t work was the possibility of losing informative data since some features with high missingness were dropped. Also, our imputation methods might not have fully captured complex patterns.

In our feature engineering steps, we applied log transformations and created features that would capture non-linear relationships and improve the predictive accuracy of our regression model. Log transformations were applied to some features that were right-skewed, such as ‘calculatedfinishedsquarefeet’ and ‘taxvaluedollarcnt’, to reduce extreme values and make the distributions closer to normal. Ratio features such as ‘bath_per_sqft’ were created to represent the number of bathrooms relative to the size of the property. Interaction features were also created, such as ‘bath_product’, which is ‘calculatedbathnbr’ multiplied by ‘bathroomcnt’, and ‘log_room_size’, which is ‘bedroomcnt’ multiplied by ‘calculatedfinishedsquarefeet’ and then logged. These were added to capture complex relationships between features and to highlight potential effects when combined. A polynomial feature, ‘age_squared’, was created by

subtracting the year 2025 from 'yearbuilt' and then raising the result to the power of two. This was added to represent non-linear effects between a property's age and its value. These newly engineered features were made to effectively uncover underlying patterns and capture non-linear relationships. This can help models to fit better on complex datasets, lower potential bias, and improve the prediction accuracy. The logged features and some of the combined features worked because they got a higher F-score and correlation score during the evaluation. In a later feature selection step, these features are also selected by the three regression models we used. Some of the combined features didn't work because they had a weak correlation and a lower F-score during the evaluation.

For the model selection process, we decided to use Gradient Boosting Regressor, Random Forest, and Decision Tree Regressor. These models were chosen because they can handle both numeric and categorical data, and can capture non-linear relationships and patterns between variables. We replaced One-Hot Encoding with LabelEncoder to encode our categorical features, and this method is better suited for tree-based regression models, which is also why we selected these models. To evaluate generalization, we used RepeatedKFold cross-validation with five repeats and five folds along with MAE as our scoring metric. From our feature engineering process, we decided to use three new features, 'log_sqft', 'bath_product', and 'age_squared', because they were considered the most useful based on the evaluation we conducted using F-scores and correlation. This was done to improve model performance. Additionally, feature selection using feature importance, forward, and backward feature selection was conducted during this phase to identify the subset of features that performed the best. To support this analysis, plots were generated depicting the change in CV scores during forward and backward feature selection across the three regression models. Hyperparameters for each model

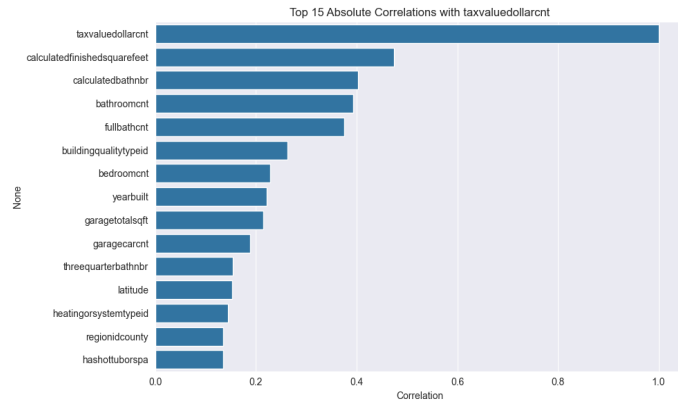
were tuned using GridSearchCV with the following parameters: `n_estimators`, `max_depth`, `max_features`, `min_samples_leaf`, and `min_samples_split`. We handled tuning by using broad ranges, and then narrowing the range from steps of 5 to 3 to 1, and found the best fit hyperparameters. We then evaluated the three models by comparing their MAE scores on the test set to see which model performed the best.

5. Results and Evaluation

Following our analytical framework, our exploratory data analysis revealed that our dataset contained features that were redundant or had many missing values, and that our target variable also had samples that contained missing values. Resolving these issues through the methods mentioned helped to reduce prediction errors and improve model stability. Furthermore, our investigation of feature relationships showed which features were strongly correlated to the target, `'taxvaluedollarcnt'`, and the most useful for modeling. Some of these included `'calculatedfinishedsquarefeet'`, `'calculatedbathnbr'`, `'bathroomcnt'`, and `'fullbathcnt'`, with correlation coefficients ranging from 0.37 to 0.47 (Figure 1). Other features showed moderate correlation, such as `'buildingqualitytypeid'`, `'bedroomcnt'`, and `'yearbuilt'`. Negatively correlated features were close to zero, suggesting that most of the features are either positively correlated or have no linear relationship with the target. These insights helped influence our feature selection and engineering, enabling our model to focus on the most essential features and improve its predictive performance.

Figure 1.

Top 15 Absolute Correlations with taxvaluedollarcnt



Note. The bar chart displays the top 15 features from the dataset that are strongly correlated to the target variable, ‘taxvaluedollarcnt’, measured by their absolute Pearson correlation coefficient. Values closer to 1 indicate a stronger correlation.

Building on these insights, we created new features in our feature engineering process. We found that ‘log_sqft’, ‘bath_product’, ‘log_room_size’, and ‘age_squared’ were the most valuable predictors, as they showed strong correlations with the target. These correlation coefficients were ~0.40, 0.33, 0.32, and -0.23, respectively. Also, these features showed high F-scores of up to 13,787. Although the correlation coefficient of ‘age-squared’ is negative, it is still vital in this context because it has a high F-score of 4,177 and captures a non-linear relationship between property age and value. It shows that as property age increases, the property value decreases, and that decrease will continue to accelerate for older properties.

With this new analysis, in our modeling process, we selected the following models: GradientBoostingRegressor, Decision Tree Regression, and Random Forest. In our establishing baseline step, we used our clean dataset without our engineered features. It showed that the Random Forest model had the lowest mean CV MAE of 134,894, followed by the GradientBoostingRegressor with 137,948, while the Decision Tree model performed the worst with 174,731 (Table 1).

Table 1

Baseline Model Performance (MAE metrics)

Model	Mean CV MAE	Standard Deviation CV MAE	Train MAE
Decision Tree Regression	174,731.68	1,218.34	2,484.98
Random Forest Regression	134,894.31	941.77	50,810.35
Gradient Boosting Regressor	137,948.22	1,068.54	136,713.08

Note. The table shows baseline performance of the three selected regression models using mean absolute error as the scoring metric. The mean and standard deviation CV MAE are from cross-validation, and the train MAE shows performance on the training set.

After applying feature engineering and specifically using the following variables:

‘log_sqft’, ‘bath_product’, and ‘age_squared’, the models performed similarly to their baseline.

For the GBR and Random Forest models, their mean CV MAE slightly increased, with Random Forest continuing to outperform. On the other hand, the Decision Tree model showed a slight decrease, but remained the least accurate model. This revealed that the newly engineered features added some predictive information to these models, but did not significantly improve model performance. With feature selection, we found that backward feature selection improved the performance of the GBR model, with its mean CV MAE being ~137,851 (Table 2). This shows that reducing features deemed less significant enhanced the model’s ability to prioritize the most relevant features. Also, our three engineered features were retained, indicating that these variables were meaningful and contributed to the predictive performance. For the Decision Tree model, forward feature selection improved the model, with a mean CV MAE of 150,273. This shows that adding features one at a time helped the model identify the features that were most informative to improve predictive accuracy. Also, none of the engineered features were retained, suggesting that these variables did not provide any additional value to the predictive performance. For the Random Forest model, feature selection with feature importance was

applied, and this revealed a mean CV MAE of ~134,906. All three engineered features were retained as well, which indicates that they contributed to the predictive value of the model.

Table 2

Model Performance After Feature Selection (MAE metrics)

Model	Feature Selection Method	Mean CV MAE	Standard Deviation CV MAE
Decision Tree Regression	Forward Feature Selection	150,273.15	1,120.08
Random Forest Regression	Feature Importance	134,905.94	1,083.23
Gradient Boosting Regressor	Backward Feature Selection	137,850.85	1,099.86

Note. The table shows model performance after feature selection using the mean absolute error as the scoring metric.

Fine-tuning these models revealed that adjusting the hyperparameters we used improved predictive performance for all three models. For the GBR model, tuning resulted in a mean CV MAE of ~134,439 (Table 3). The Decision Tree and Random Forest models also improved, showing a mean CV MAE of 146,345 and ~133,836, respectively. Random Forest continued to show the strongest performance, suggesting that feature selection and hyperparameter tuning helped the model to capture the most informative features for accurately predicting property valuations. Overfitting was observed in all the models, so we adjusted specific parameters, such as `max_depth` and `min_samples_leaf`, to reduce it. Lastly, a held-out test set was used on these three models to evaluate their performance on unseen data. We found that GBR slightly outperformed Random Forest as it achieved the lowest test MAE of ~136,208. The Random Forest model resulted in a test MAE of ~136,619, while the Decision Tree model performed the worst at 147,113.

Table 3*Model Performance on Cross-Validation and Held-Out Test Set*

Model	Mean CV MAE	Standard Deviation CV MAE	Test MAE
Decision Tree Regression	143,918.39	1,221.61	147113.16
Random Forest Regression	132,865.62	1,005.54	136,618.75
Gradient Boosting Regressor	134,438.93	971.60	136,207.57

Note. The table shows cross-validation and held-out test set results, using the mean absolute error as the scoring metric.

After comparing these results of the three models, we selected GradientBoostingRegressor as our final model, as it achieved the lowest MAE score on the test set from the RepeatedKfold cross-validation. Also, it had the lowest standard deviation CV MAE of ~972 when compared to the other models. This indicates that GBR made accurate predictions and maintained a consistent performance across folds. Due to the model being an ensemble method, it was able to capture complex relationships by learning from its errors. Also, the model retained all three of the engineered features, showing that it used the most valuable features to make accurate predictions on property valuation without fitting too closely to the training data. This allowed it to maintain good generalization and perform reliably on unseen data.

Specific patterns of error we noticed were that GBR had the longest training time compared to the other models, it slightly overfitted with the initial best hyperparameters, and that its test MAE score was higher than the train MAE score. However, by reducing the max_depth and increasing the min_sample_leaf slightly, it performed significantly better than before.

During the fine-tuning, the model initially got a lower MAE score on the training set but a higher MAE on the test set, which is a sign of overfitting. This occurred because the GridSearchCV tuned too narrowly for the hyperparameters. To address this, `min_samples_leaf` was increased and `max_depth` was reduced, which improved the generalization.

In future iterations, it is worth improving the deployment of the pipeline because it could save more time when building more models instead of defining each model one by one.

6. Conclusion:

We were able to predict `taxvaluedollarcnt` with relatively low absolute error after careful cleaning, imputation, encoding, and targeted feature engineering such as log transforms and derived ratios. Among the models we tested (GradientBoostingRegressor, Random Forest, and Decision Tree), GradientBoostingRegressor performed best on the held-out test set (MAE $\approx$ 136,207), slightly ahead of Random Forest ($\approx$ 136,618) and Decision Tree ($\approx$ 147,113). Simple regularization adjustments, such as lowering `max_depth` and increasing `min_samples_leaf`, improved generalization and stabilized errors across value ranges. The engineered features were consistently retained by the tree-based models. These results indicate that more reliable value estimates support pricing, portfolio planning, and customer tools. They also highlight the features that matter most for prediction to guide data quality priorities and feature engineering. However, our analysis had limitations. These included substantial missingness, possible label noise in assessor values, use of a subset that may not have captured all geographies or market conditions, and the absence of time-aware validation. Future work should consider incorporating geospatial and temporal signals, experimenting with LightGBM or CatBoost, and using time or location-based cross-validation. Also, quantifying uncertainty with prediction intervals or quantile loss, applying SHAP for systematic interpretability, strengthening imputation with

model-based methods, and considering ensembling to further reduce variance could further enhance our model's accuracy and robustness.